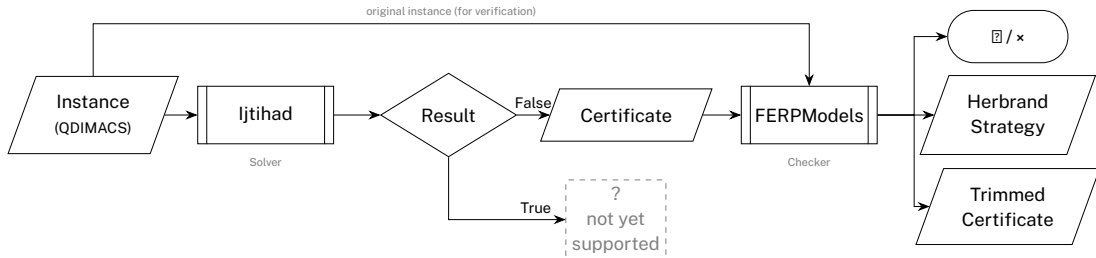


FERPModels Certification Pipeline



What makes $\exists$ -Expansion Hard to Certify?

Quantifier	$\Pi \forall \mathbf{0} u . \phi$	$\Pi \exists \mathbf{0} e . \phi$
Expansion	$\Pi . \phi[u/\boxed{?}] \wedge \phi[u/\boxed{?}]$	$\Pi . \phi[e/\boxed{?}] \vee \phi[e/\boxed{?}]$
Structure	CNF	non-CNF

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Structure	CNF	non-CNF
Check	Satisfiability	Validity

$\forall$ Expansion Proofs

$\exists x \forall a \exists y.$

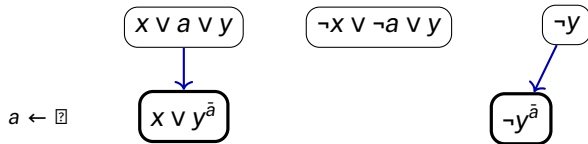
$x \vee a \vee y$

$\neg x \vee \neg a \vee y$

$\neg y$

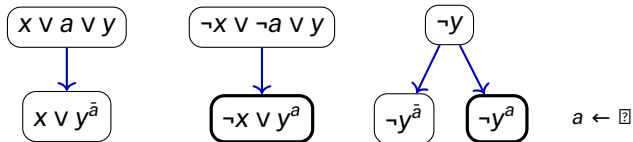
$\forall$ Expansion Proofs

$\exists x \forall a \exists y.$

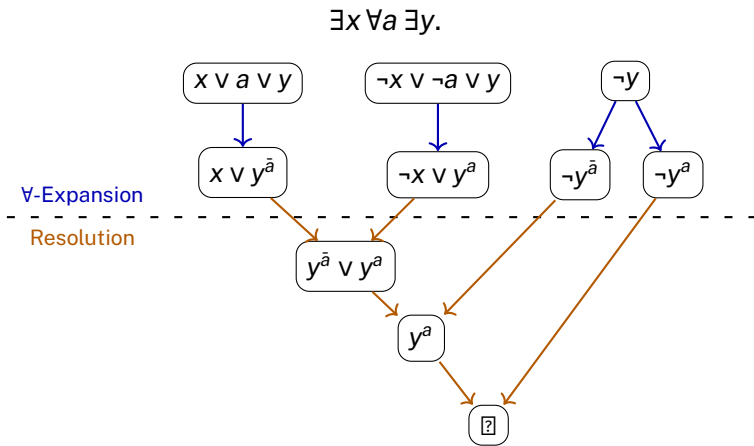


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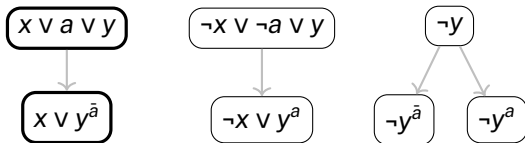


$\forall$ Expansion Proofs



$\forall$ Expansion Proofs

$\exists x \forall a \exists y.$



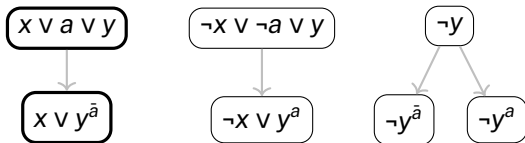
Checking $(x \vee y^a)$

$(x \vee y^{\bar{a}})$

$(x \vee a \vee y)$

$\forall$ Expansion Proofs

$\exists x \forall a \exists y.$



Checking $(x \vee y^a)$

$(x \vee y^{\bar{a}}) \xrightarrow{\text{strip \& store annotation}} (x \vee y), a \leftarrow ? \quad (x \vee a \vee y)$

$\forall$ Expansion Proofs

$\exists x \forall a \exists y.$

$x \vee a \vee y$

$x \vee y^{\bar{a}}$

$\neg x \vee \neg a \vee y$

$\neg x \vee y^a$

$\neg y$

$\neg y^{\bar{a}}$

$\neg y^a$

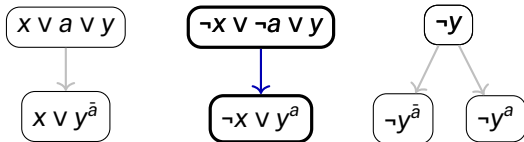
Checking $(x \vee y^a)$

$(x \vee y^{\bar{a}}) \xrightarrow{\text{strip \& store annotation}} (x \vee y), a \leftarrow ?$

$(x \vee a \vee y)[a \leftarrow ?] = (x \vee y)$

$\forall$ Expansion Proofs

$\exists x \forall a \exists y.$



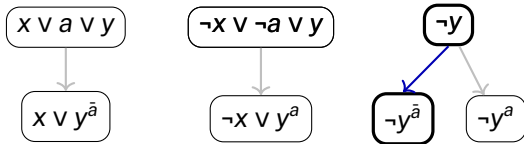
Checking $(\neg x \vee y^a)$

$(\neg x \vee y^a) \xrightarrow{\text{strip \& store annotation}} (\neg x \vee y), a \leftarrow ?$

$(\neg x \vee \neg a \vee y)[a \leftarrow ?] = (\neg x \vee y)$

$\forall$ Expansion Proofs

$\exists x \forall a \exists y.$



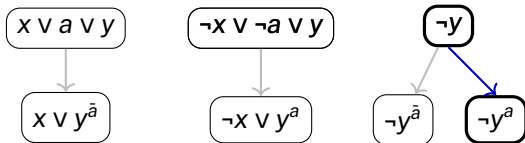
Checking $(\neg y^{\bar{a}})$

$(\neg y^{\bar{a}}) \xrightarrow{\text{strip \& store annotation}} (\neg y), a \leftarrow ?$

$(\neg y)[a \leftarrow ?] = (\neg y)$

$\forall$ Expansion Proofs

$\exists x \forall a \exists y.$



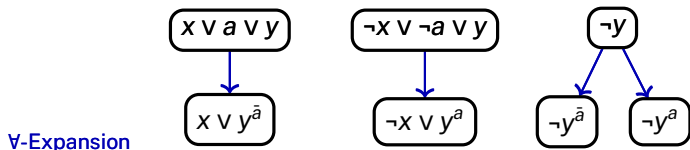
Checking $(\neg y^a)$

$(\neg y^a) \xrightarrow{\text{strip \& store annotation}} (\neg y), a \leftarrow \boxed{?}$

$(\neg y)[a \leftarrow \boxed{?}] = (\neg y)$

Why $\forall$ -Expansion Proofs Work Well

$\exists x \forall a \exists y.$

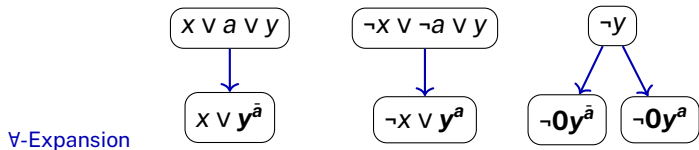


Clause Correspondence

Each expansion clause maps to exactly one matrix clause
the checker knows unambiguously which clause to verify against

Why $\forall$ -Expansion Proofs Work Well

$$\exists x \forall a \exists y.$$



Assignment Recovery

Universal reduction guarantees the innermost quantifier is existential.
The instantiating assignment is always recoverable from annotations.

Universal Expansion

$$\forall x. \phi \equiv \phi[x/\top] \wedge \phi[x/\bot]$$

Existential Expansion

$$\exists x. \phi \equiv \phi[x/\top] \vee \phi[x/\bot]$$

Universal Expansion

$$\forall x. \phi \equiv \phi[x/\top] \wedge \phi[x/\bot]$$

$$\Psi = \Pi_1 \forall u \Pi_2. \phi$$

Existential Expansion

$$\exists x. \phi \equiv \phi[x/\top] \vee \phi[x/\bot]$$

$$\Psi = \Pi_1 \exists e \Pi_2. \phi$$

Universal Expansion

$$\forall x. \phi \equiv \phi[x/\top] \wedge \phi[x/\bot]$$

$$\Psi = \Pi_1 \left(\Pi_2^{\bar{u}}. \phi^{\bar{u}} \wedge \Pi_2^u. \phi^u \right)$$

Existential Expansion

$$\exists x. \phi \equiv \phi[x/\top] \vee \phi[x/\bot]$$

$$\Psi = \Pi_1 \left(\Pi_2^{\bar{e}}. \phi^{\bar{e}} \vee \Pi_2^e. \phi^e \right)$$

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$$\alpha : U_\Pi \rightarrow \mathbb{B}, \quad A \subseteq \{0, 1\}^{U_\Pi}$$

$$\exists \underbrace{\bigwedge_{\alpha \in A} \phi^\alpha}_{\text{conjunction of CNFs}} = \top \implies \Psi \text{ false}$$

Existential Expansion

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Existential Expansion

$$\exists x. \phi \equiv \phi[x/\top] \vee \phi[x/\bot]$$

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$$\sigma : E_\Pi \rightarrow \mathbb{B}, \quad S \subseteq \{0, 1\}^{E_\Pi}$$

$$\neg(\forall \underbrace{\bigvee_{\sigma \in S} \phi^\sigma}_{\text{disjunction of CNFs}}) = \top \implies \Psi \text{ true}$$

Universal Expansion

$$\forall x. \phi \equiv \phi[x/\top] \wedge \phi[x/\bot]$$

$$\Psi = \Pi_1 \Pi_2^{\bar{u}} \Pi_2^u. \phi^{\bar{u}} \wedge \phi^u$$

$$\alpha : U_{\Pi} \rightarrow \mathbb{B}, \quad A \subseteq \{0, 1\}^{U_{\Pi}}$$

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Existential Expansion

$$\exists x. \phi \equiv \phi[x/\top] \vee \phi[x/\bot]$$

$$\Psi = \Pi_1 \Pi_2^{\bar{e}} \Pi_2^e. \phi^{\bar{e}} \vee \phi^e$$

$$\sigma : E_{\Pi} \rightarrow \mathbb{B}, \quad S \subseteq \{0, 1\}^{E_{\Pi}}$$

$$\exists \underbrace{\bigwedge_{\sigma \in S} \neg \phi^{\sigma}}_{\text{conjunction of DNFs}} = \top \implies \Psi \text{ true}$$

Expansion Proofs

$$\forall a \exists x \forall b \exists y.$$

$$\neg y \vee \neg a \vee b$$

$$y \vee a$$

$$y \vee x \vee \neg b$$

Expansion Proofs

$$\forall a \exists x \forall b \exists y.$$

$$\neg y \vee \neg a \vee b$$

$$y \vee a$$

$$y \vee x \vee \neg b$$

$$\neg \phi^{\bar{x}, \bar{y}}$$

$$x \leftarrow ?, y \leftarrow ?$$

Expansion Proofs

$$\forall a \exists x \forall b \exists y.$$

$$\neg y \vee \neg a \vee b$$

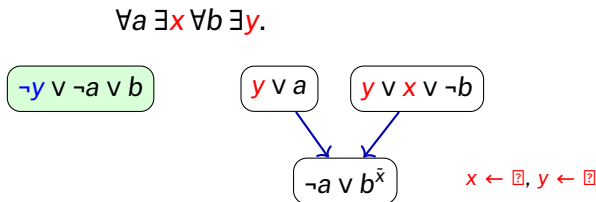
$$y \vee a$$

$$y \vee x \vee \neg b$$

$$\neg((a) \wedge (\neg b^{\bar{x}}))$$

$$x \leftarrow ?, y \leftarrow ?$$

∃ Expansion Proofs



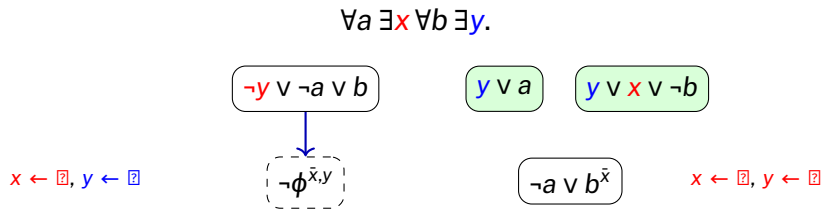
Reference Clause

C_1 is satisfied.

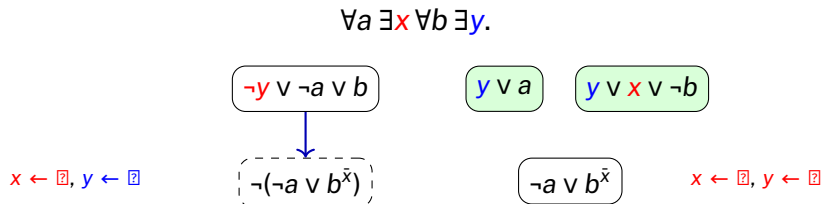
C_2 and C_3 survive under $\phi[x \leftarrow ?, y \leftarrow ?]$ as unit cubes.

Encoded as a single reference clause $(\neg a \vee b^{\bar{x}})$ after negating.

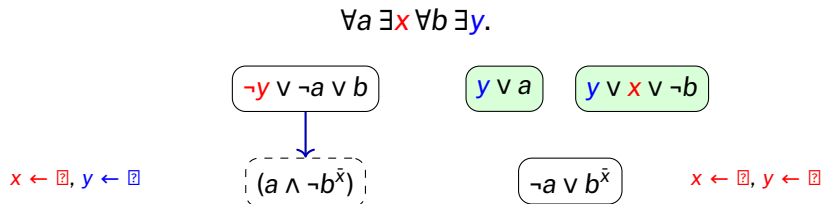
$\exists$ Expansion Proofs



Expansion Proofs



Expansion Proofs



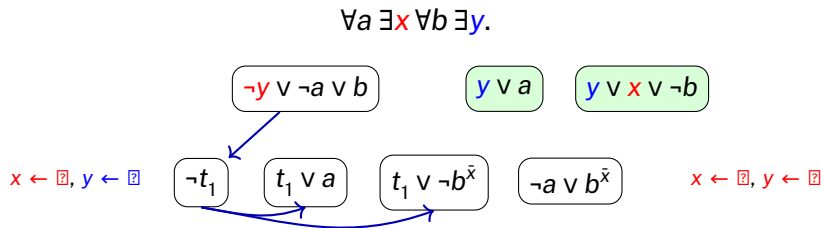
Plaisted-Greenbaum for C_1

Introduce t_1 for $(a \wedge \neg b^{\bar{x}})$: $t_1 \Rightarrow (a \wedge \neg b^{\bar{x}})$

Reference clause: $(\neg t_1)$

Definition clauses: $(t_1 \vee a) \wedge (t_1 \vee \neg b^{\bar{x}})$

Expansion Proofs



Plaisted-Greenbaum for C_1

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Reference clause: $(\neg t_1)$

Definition clauses: $(t_1 \vee a) \wedge (t_1 \vee \neg b^{\bar{x}})$

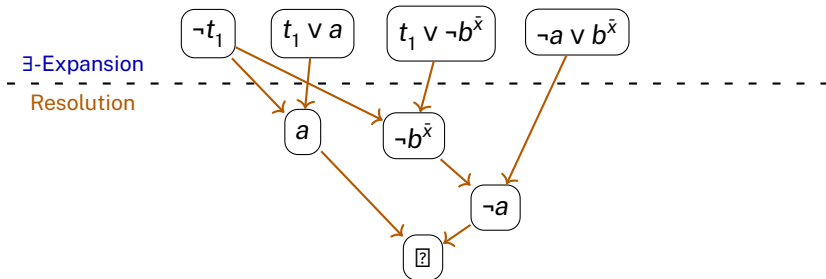
$\exists$ Expansion Proofs

$$\forall a \exists x \forall b \exists y.$$

$$\neg y \vee \neg a \vee b$$

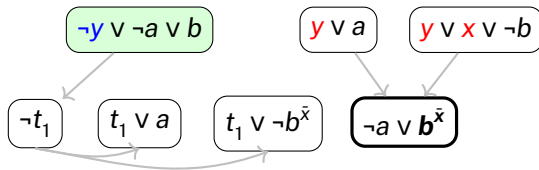
$$y \vee a$$

$$y \vee x \vee \neg b$$



Checking $\exists$ Expansion Proofs

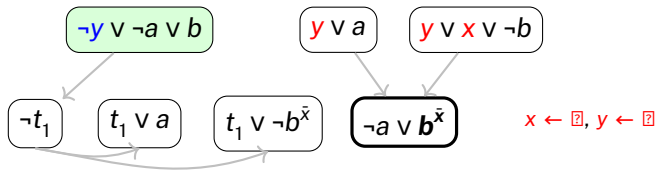
$\forall a \exists x \forall b \exists y.$



Checking $b^{\bar{x}}$ represents $\neg(y \vee x \vee \neg b)$
 $(b^{\bar{x}}) \quad \neg C_3 = (\neg y \wedge \neg x \wedge b)$

Checking $\exists$ Expansion Proofs

$\forall a \exists x \forall b \exists y.$



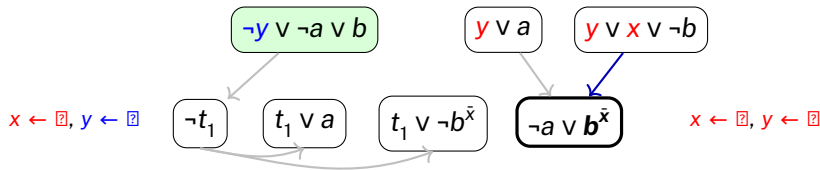
Checking $b^{\bar{x}}$ represents $\neg(y \vee x \vee \neg b)$

$(b^{\bar{x}}) \xrightarrow{\text{strip \& store annotation}} (b), x \leftarrow ?$

$\neg C_3[x \leftarrow ?] = (\neg y \wedge b)$

Checking $\exists$ Expansion Proofs

$\forall a \exists x \forall b \exists y.$



Checking $b^{\bar{x}}$ represents $\neg(y \vee x \vee \neg b)$

$(b^{\bar{x}}) \xrightarrow{\text{strip \& store annotation}} (b), x \leftarrow ?$

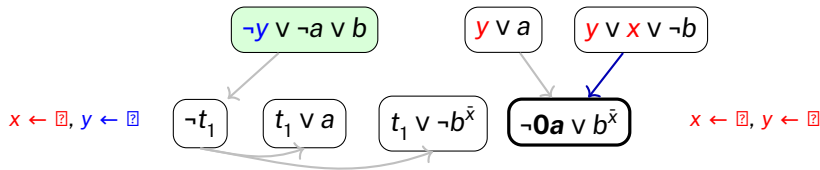
$\neg C_3[x \leftarrow ?] = (\neg y \wedge b)$

$\neg y$ unknown from annotation alone

$\neg C_3[x \leftarrow ?, y \leftarrow ?] = (b) \sqcap$

Checking $\exists$ Expansion Proofs

$\forall a \exists x \forall b \exists y.$



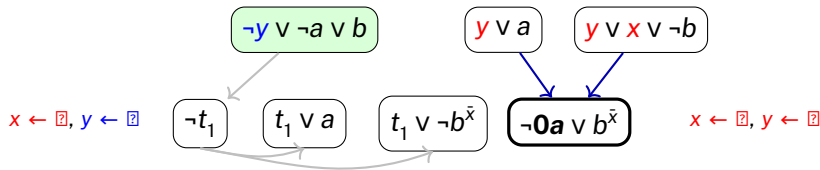
Checking $\neg a$ represents $\neg(y \vee a)$

$(\neg a)$

$\neg C_2 = (\neg y \wedge \neg a)$

Checking $\exists$ Expansion Proofs

$\forall a \exists x \forall b \exists y.$



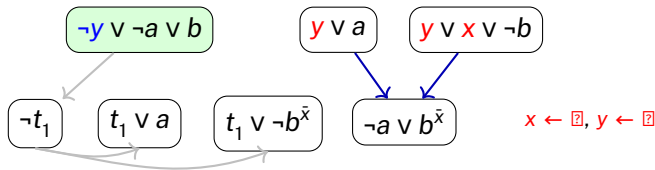
Checking $\neg a$ represents $\neg(y \vee a)$

strip & store
annotation
 $(\neg a) \xrightarrow{\quad} (\neg a), \emptyset$

$\neg C_2[x \leftarrow ?, y \leftarrow ?] = (\neg a) \sqcap$

Checking $\exists$ Expansion Proofs

$\forall a \exists x \forall b \exists y.$



Validity Check for $\sigma = \{\bar{x}, \bar{y}\}$

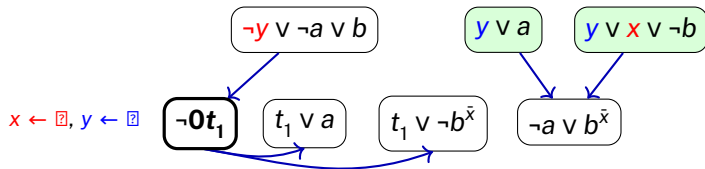
$C_1[x \leftarrow \text{?}, y \leftarrow \text{?}] = \text{?} \models$ satisfied — correctly eliminated

$C_2[x \leftarrow \text{?}, y \leftarrow \text{?}] = (a) \not\models$ unsatisfied — correctly survives

$C_3[x \leftarrow \text{?}, y \leftarrow \text{?}] = (\neg b) \not\models$ unsatisfied — correctly survives

Checking $\exists$ Expansion Proofs

$\forall a \exists x \forall b \exists y.$



Checking $\neg 0t_1$ represents $\neg(\neg y \vee \neg a \vee b)$

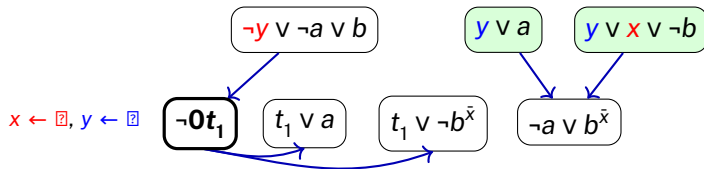
Retrieve cube from definition clauses: $(a \wedge \neg b^{\bar{x}})$

$(a \wedge \neg b^{\bar{x}})$

$\neg C_1 = (y \wedge a \wedge \neg b)$

Checking $\exists$ Expansion Proofs

$\forall a \exists x \forall b \exists y.$



Checking $\neg t_1$ represents $\neg(\neg y \vee \neg a \vee b)$

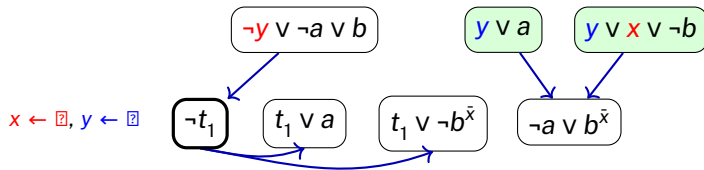
Retrieve cube from definition clauses: $(a \wedge \neg b^{\bar{x}})$

$(a \wedge \neg b^{\bar{x}}) \xrightarrow{\text{strip \& store annotation}} (a \wedge \neg b), x \leftarrow \text{?}$

$\neg C_1[x \leftarrow \text{?}, y \leftarrow \text{?}] = (a \wedge \neg b) \sqcap$

Checking $\exists$ Expansion Proofs

$\forall a \exists x \forall b \exists y.$



Validity Check for $\sigma = \{\bar{x}, y\}$

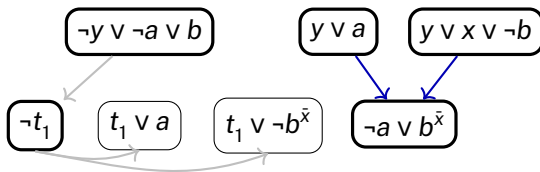
$C_1[x \leftarrow [?], y \leftarrow [?]] = (\neg a \vee b) \sqcap$ unsatisfied — correctly survives

$C_2[x \leftarrow [?], y \leftarrow [?]] = [?] \sqcap$ satisfied — correctly eliminated

$C_3[x \leftarrow [?], y \leftarrow [?]] = [?] \sqcap$ satisfied — correctly eliminated

$\exists$ -Expansion Certification: Summary

$$\forall a \exists x \forall b \exists y.$$

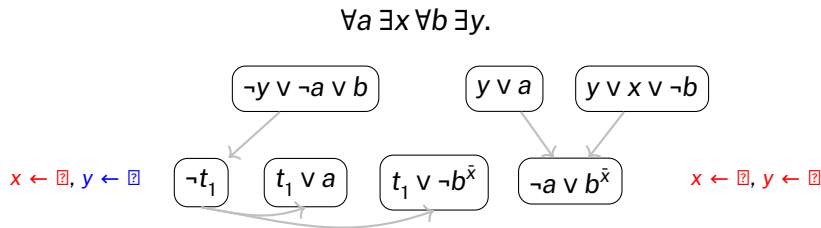


Broken Clause Correspondence

$(\neg a \vee b^{\bar{x}})$ maps to both C_2 and C_3 .

Each literal in a reference clause is a representative of exactly one matrix clause.

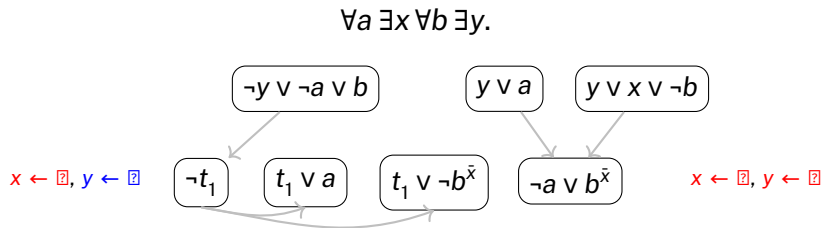
$\exists$ -Expansion Certification: Summary



Incomplete Annotation

y is innermost and annotates nothing on expansion
the complete σ must be explicitly recorded in the proof for each reference clause.

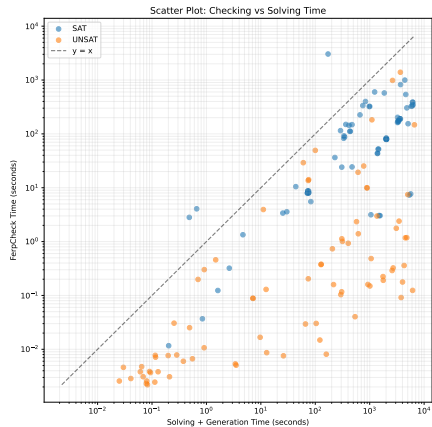
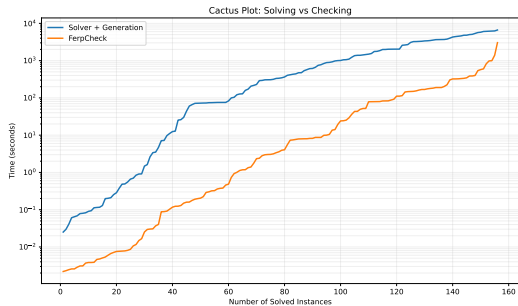
$\exists$ -Expansion Certification: Summary



Validity Check

the complete σ recorded in the proof is used to verify that eliminated clauses are genuinely satisfied and surviving clauses are correctly unsatisfied

Preliminary Experiments



Future Work

- Proof trimming for $\exists$ -expansion proofs.

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- Skolem function extraction from verified $\exists$ -expansion proofs.

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- Proof trimming for $\exists$ -expansion proofs.
- Skolem function extraction from verified $\exists$ -expansion proofs.
- Extensive experiments comparing to other proof system implementations (QBFCert, FERAT, FERPModels, CAQE, etc.).

Temporary page!

$\text{\LaTeX}$ was unable to guess the total number of pages correctly. As there was unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away because $\text{\LaTeX}$ now knows how many pages to expect for this document.